1 Solution — GIAM 2.2 Problem 8

1.1 Problem

Find the contrapositives of the following sentences.

- *(a)* If you can't do the time, don't do the crime.
- *(b)* If you do well in school, you'll get a good job.
- (c) If you wish others to treat you in a certain way, you must treat others in that fashion.
- *(d)* If it's raining, there must be clouds.
- *(e)* If $a_n \leq b_n$ for all n and $\sum_{n=0}^{\infty} b_n$ is a convergent series, then $\sum_{n=0}^{\infty} a_n$ is a convergent series.

1.2 The Rule

The contrapositive of $P \Rightarrow Q$ is $\neg Q \Rightarrow \neg P$ — **swap** the two parts and **negate** both. By Problem 2's rewrite the two are logically equivalent, so a statement and its contrapositive always have the same truth value.

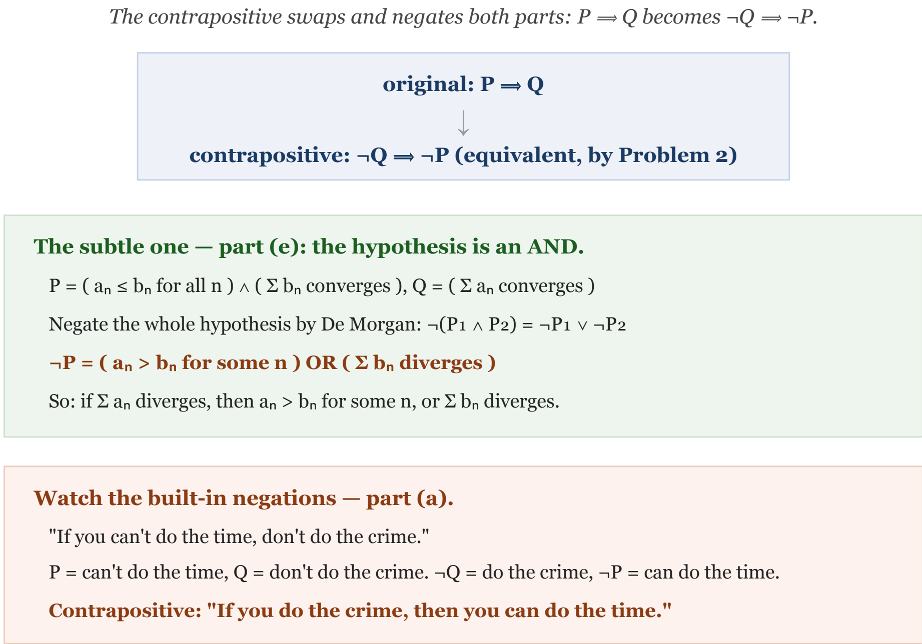

Figure 1.1: The contrapositive swaps and negates both parts: P implies Q becomes not-Q implies not-P, and the two are equivalent. The subtle case, part (e): the hypothesis is an AND, P = (a-n at most b-n for all n) and (sum of b-n converges), Q = (sum of a-n converges). Negate the whole hypothesis by De Morgan: not(P1 and P2) = not-P1 or not-P2, giving 'a-n greater than b-n for some n, OR sum of b-n diverges'. So: if sum of a-n diverges, then a-n exceeds b-n for some n, or sum of b-n diverges. Part (a) warns to watch built-in negations: with P = can't do the time and Q = don't do the crime, not-Q = do the crime and not-P = can do the time, so the contrapositive is 'If you do the crime, then you can do the time.'

1.3 The Contrapositives

(a) Original: P = "you can't do the time," Q = "don't do the crime." Negating each undoes the built-in "can't"/"don't":

$$\boxed{\text{If you } \textit{do} \text{ the crime, then you } \textit{can} \text{ do the time.}}$$

(b) P = "you do well in school," Q = "you'll get a good job":

$$\boxed{\text{If you don't get a good job, then you didn't do well in school.}}$$

(c) P = "you wish others to treat you in a certain way," Q = "you must treat others in that fashion":

$\boxed{\text{If you don't treat others in a certain fashion, then you don't wish to be treated that way.}}$

(d) $P =$ "it's raining," $Q =$ "there are clouds":

$$\boxed{\text{If there are no clouds, then it isn't raining.}}$$

(e) Here the hypothesis is a **conjunction**: $P = \big(a_n \le b_n \text{ for all } n\big) \wedge \big(\sum b_n \text{ converges}\big)$, and $Q = \big(\sum a_n \text{ converges}\big)$. Negating P requires De Morgan, turning the "and" into an "or":

$$\boxed{\begin{array}{l}\text{If } \sum_{n=0}^{\infty} a_n \text{ diverges, then either } a_n > b_n \text{ for some } n,\\ \text{or } \sum_{n=0}^{\infty} b_n \text{ diverges.}\end{array}}$$

1.4 Remark — Watch the Negations Already in the Sentence

Part (a) is the trap. Both of its clauses *already contain* a negation ("can't," "don't"), so forming the contrapositive **removes** them rather than adding more. The contrapositive of "if you can't do the time, don't do the crime" is *not* "if you can do the crime, you can't do the time" — one must negate each clause as a whole:

$$\neg Q = \neg(\text{don't do the crime}) = \text{do the crime},$$
$$\neg P = \neg(\text{can't do the time}) = \text{can do the time}.$$

The double negations cancel, giving the clean positive statement in the box. Whenever a clause is already negative, negating it should *simplify* the English, not pile on another "not."

1.5 Remark — Part (e): Negating a Compound Hypothesis

Part (e) is the mathematically substantial one, because its hypothesis has two parts joined by "and." De Morgan's law says

$$\neg(P_1 \wedge P_2) \;\equiv\; \neg P_1 \vee \neg P_2,$$

so the negated hypothesis becomes a *disjunction*: at least one of the two conditions fails. Note also the careful negation of the quantified clause — the opposite of "$a_n \le b_n$ **for**

all n" is "$a_n > b_n$ **for some** n," not "$a_n > b_n$ for all n." (Negating a universal gives an existential — the topic Chapter 2 takes up next.) Both subtleties must be handled for the contrapositive to be correct.

1.6 Remark — Why Contrapositives Are Useful in Proofs

The contrapositive is not just a grammatical exercise: it is a **proof strategy**. Since $P \Rightarrow Q$ and $\neg Q \Rightarrow \neg P$ are equivalent, one may prove either. Often the contrapositive is dramatically easier, because its hypothesis gives you something concrete to work with. Statement (e) is a perfect example: it is the **comparison test** from calculus, and its contrapositive is the form actually used in practice — to show a series *diverges*, you compare it to a smaller divergent series. Similarly, Problem 2.1.3's parity lemma ("if x^2 is even then x is even") was proved by its contrapositive, because "x is odd" hands you the formula $x = 2k + 1$ while "x^2 is even" hands you nothing usable.

1.7 Remark — Contrapositive vs. Converse vs. Inverse

Three transformations of $P \Rightarrow Q$ are easily confused, and only one preserves truth:

name	form	equivalent to original?
contrapositive	$\neg Q \Rightarrow \neg P$	**yes ✓**
converse	$Q \Rightarrow P$	no
inverse	$\neg P \Rightarrow \neg Q$	no

Table 1.1

Statement (d) makes the difference vivid. The original ("if raining, then clouds") is true; its contrapositive ("if no clouds, then not raining") is also true. But the *converse* ("if there are clouds, then it's raining") is plainly **false** — cloudy days are often dry. The converse and inverse are equivalent to each other (each is the contrapositive of the other), but neither follows from the original. Confusing a statement with its converse is among the most common reasoning errors, in mathematics and in daily life alike.

1.8 Remark — The Golden Rule, Contraposed

Statement (c) is a rendering of the **Golden Rule**, and contraposing it gives a striking rhetorical shift. The original is aspirational — "treat others as you wish to be treated" — while the contrapositive reads as a diagnosis: *if you are not treating others well, then you do not really wish to be treated well yourself.* Logically the two say precisely the same thing, yet they land very differently on the ear. This is a nice reminder that logical equivalence is about *truth conditions* only; emphasis, tone, and persuasive force are not preserved by the transformation, even though truth value always is.